
Basis Fidelity vs. Average Fidelity

What Pedersen, Møller, Mølmer (2007) actually tells you

Reading note¹

¹Pedersen et al., Phys. Lett. A 367 (2007) 47–51

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ABSTRACT

Pedersen, Møller, and Mølmer derive a compact closed form for the average fidelity of a quantum operation, averaged uniformly over pure input states. A natural practical question is whether knowing only the *basis fidelities* (the fidelity when each computational-basis state is used as input) is enough to recover the average. **It is not.** The average fidelity depends on a bilinear combination of Kraus-operator traces that mixes the diagonal entries across basis indices, and these cross-terms encode coherence information that basis-state measurements destroy. We make the obstruction explicit, give a minimal qubit counter-example, and explain which state sets (2-designs) do suffice.

Keywords. average fidelity · basis fidelity · Kraus operators · unitary 2-design · process tomography

1 The question in one line

Fix a target unitary U_0 on an n -dimensional Hilbert space and an actual CPTP map $\mathcal{G}$. Define the *basis fidelity* on the computational basis state $|m\rangle$:

$$F_m = \langle m | U_0^\dagger \mathcal{G}(|m\rangle\langle m|) U_0 | m \rangle. \quad (1)$$

Define the *average fidelity*:

$$\bar{F} = \int_{S^{2n-1}} \langle \psi | U_0^\dagger \mathcal{G}(|\psi\rangle\langle\psi|) U_0 | \psi \rangle dV, \quad (2)$$

with dV the normalized measure on the unit sphere. **Does the tuple $\{F_0, F_1, \dots, F_{n-1}\}$ determine $\bar{F}$?** No. The average fidelity is blind neither to populations nor to phases; basis fidelities are blind to phases. They see strictly less information.

2 The Pedersen closed form

The central identity of [1] (Theorem, Eq. 1) is that for *any* linear operator M on $\mathbb{C}^n$,

$$\int_{S^{2n-1}} |\langle \psi | M | \psi \rangle|^2 dV = \frac{1}{n(n+1)} [\text{Tr}(MM^\dagger) + |\text{Tr}(M)|^2]. \quad (1)$$

Specializing to a target unitary with Kraus operators $\{G_k\}$ for $\mathcal{G}$ and $M_k := U_0^\dagger G_k$ gives Eq. (5):

$$\bar{F} = \frac{1}{n(n+1)} \left[\sum_k \text{Tr}(M_k^\dagger M_k) + \sum_k |\text{Tr}(M_k)|^2 \right]. \quad (2)$$

Because $\mathcal{G}$ is trace-preserving, $\sum_k G_k^\dagger G_k = I$, hence $\sum_k M_k^\dagger M_k = I$ and the first sum collapses to n . All nontrivial content of $\bar{F}$ lives in the second sum:

$$\bar{F} = \frac{1}{n+1} + \frac{1}{n(n+1)} \sum_k |\text{Tr}(M_k)|^2. \quad (3)$$

In particular, the best possible value is $\bar{F} = 1$ when $\sum_k |\text{Tr}(M_k)|^2 = n^2$, the worst value a general CPTP map can reach is the depolarizing floor $\bar{F} = \frac{1}{n+1}$.

3 What a basis fidelity actually measures

Writing M_k in the computational basis with entries $(M_k)_{ij} = \langle i|M_k|j\rangle$, the basis fidelity is a sum over Kraus branches of *diagonal-entry moduli squared*:

$$F_m = \sum_k |\langle m|M_k|m\rangle|^2 = \sum_k |(M_k)_{mm}|^2. \quad (1)$$

Summing 1 over m :

$$\sum_{m=0}^{n-1} F_m = \sum_k \sum_m |(M_k)_{mm}|^2. \quad (2)$$

This is the total squared-modulus of every diagonal entry of every Kraus operator — a *diagonal* quantity.

4 Where the gap opens

Compare to what $\bar{F}$ actually needs. Expanding the trace:

$$\sum_k |\text{Tr}(M_k)|^2 = \sum_k \left| \sum_m (M_k)_{mm} \right|^2 = \underbrace{\sum_k \sum_m |(M_k)_{mm}|^2}_{= \sum_m F_m} + \underbrace{\sum_k \sum_{m \neq m'} (M_k)_{mm}^* (M_k)_{m'm'}}_{\text{cross terms}}. \quad (1)$$

The first piece is exactly the sum of basis fidelities 2. The second piece is a *phase-sensitive bilinear* in the diagonal Kraus entries. It is what you lose the instant you only look at basis populations.

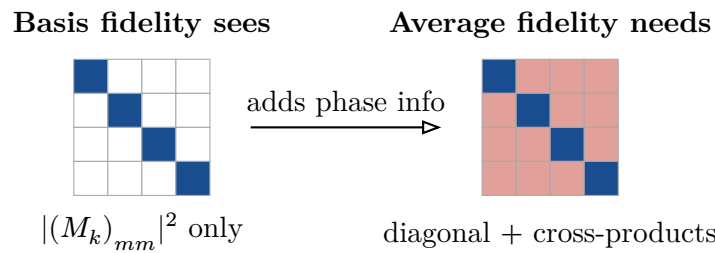


Figure 1: Schematic of information content. Basis fidelity F_m retains only the moduli of the on-diagonal Kraus entries (blue). The average fidelity additionally needs the phase-sensitive cross-products $(M_k)_{mm}^* (M_k)_{m'm'}$ for $m \neq m'$ (red). These cross-terms are *not* recoverable from $\{F_m\}$ alone.

5 A two-line counter-example

Take a qubit ($n = 2$), target $U_0 = I$, actual unitary

$$U = \text{diag}(1, e^{i\varphi}). \quad (1)$$

Then $M = U_0^\dagger U = U$ and:

- **Basis fidelities:** $F_0 = |\langle 0|M|0\rangle|^2 = 1$ and $F_1 = |\langle 1|M|1\rangle|^2 = 1$ for *every* φ .
- **Average fidelity:** $\text{Tr}(M) = 1 + e^{i\varphi}$, so $|\text{Tr}(M)|^2 = 2 + 2\cos\varphi$, and

$$\overline{F} = \frac{1}{2 \cdot 3} [2 + (2 + 2\cos\varphi)] = \frac{2 + \cos\varphi}{3}. \quad (2)$$

At $\varphi = 0$, $\overline{F} = 1$. At $\varphi = \pi$, $\overline{F} = \frac{1}{3}$ — the worst value a qubit CPTP map can give. The basis fidelities are identical in both cases. A phase-flip is invisible to population measurements in the Z basis but catastrophic for the average.

This is the same physics Pedersen et al. warn about when discussing subspace fidelities: “*phases acquired by amplitudes on the excited states cause a reduction in the fidelity, even if the final state is the correct one for all input qubit states.*” The qubit register version is just an instance of that observation.

6 What state sets are sufficient?

The computational basis is a *1-design*: it reproduces first-order moments of Haar-random states, $\frac{1}{n} \sum_m |m\rangle\langle m| = \frac{I}{n}$. The average fidelity is *quadratic* in the input state, so it needs *second-order* Haar moments — a *2-design* [2,3].

- **Qubit.** Four states of a regular tetrahedron on the Bloch sphere form a 2-design; so do the six axis states $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle, |+i\rangle, |-i\rangle\}$ (the Bowdrey construction [2]). Either set determines $\overline{F}$.
- **Qudit.** A complete set of mutually unbiased bases (when it exists), any Clifford orbit, or any state 2-design [3].
- **Arbitrary.** Full process tomography [4], then plug Kraus operators into 2.

The basis is *strictly weaker* than any of these. It is a 1-design precisely, so it pins down the diagonal of the process matrix χ but leaves the off-diagonal entries — the phase-sensitive part — undetermined.

7 Practical consequences for Rydberg gate simulation

In the context of this project — simulating Rydberg two-qubit gates via master-equation or Monte Carlo Wave Function solves — the useful reading of [1] is:

- (i) **If you have the full actual evolution** (U or the Kraus set $\{G_k\}$), use 2 directly. One trace per Kraus operator, sum, done. No sampling, no tomography. This is the numerically cheap and correct path.
- (ii) **If you only read out computational-basis populations** (e.g., only the CCX or CNOT truth-table populations $|\langle m_{\text{out}}|U|m_{\text{in}}\rangle|^2$), you are measuring a *phase-blind proxy*. It is useful for debugging, but *cannot* certify the gate’s average fidelity. Two runs with identical truth-table populations can differ in $\overline{F}$ by as much as $\frac{2}{n+1}$.
- (iii) **If you want a cheap experimental proxy for $\overline{F}$** , pick a 2-design — for a qubit the tetrahedron; for two qubits the 20-state set of Bowdrey or any Clifford 2-design — and average the state fidelities over it.
- (iv) **Subspace fidelity** (Pedersen Eq. 3) is the right tool when auxiliary Rydberg levels carry leaked population or acquired phases. The relevant matrix is $M_{\text{rel}} = PU_0^\dagger UP$ with P the projector onto the qubit subspace, and the same Haar-average theorem applies on $S^{2n_{\text{rel}}-1}$.

8 Summary box

Claim. Basis fidelities $\{F_m\}_{m=0}^{n-1}$ do not determine the average fidelity $\bar{F}$.

Reason. $\bar{F} = \frac{1}{n+1} + (n(n+1))^{-1} \sum_k |\text{Tr}(M_k)|^2$. Expanding $|\text{Tr}(M_k)|^2$ yields $\sum_m |(M_k)_{mm}|^2 + \sum_{m \neq m'} (M_k)_{mm}^* (M_k)_{m'm'}$. The first piece equals $\sum_m F_m$; the second is a phase-sensitive cross-term that no basis-population measurement can reveal.

Minimal witness. $U = \text{diag}(1, e^{i\varphi})$ on a qubit: $F_0 = F_1 = 1$ for all φ , but $\bar{F} = \frac{2+\cos \varphi}{3} \in [\frac{1}{3}, 1]$.

Fix. Use the closed-form 2 when you have Kraus data; otherwise average the state fidelity over a state 2-design.

References

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